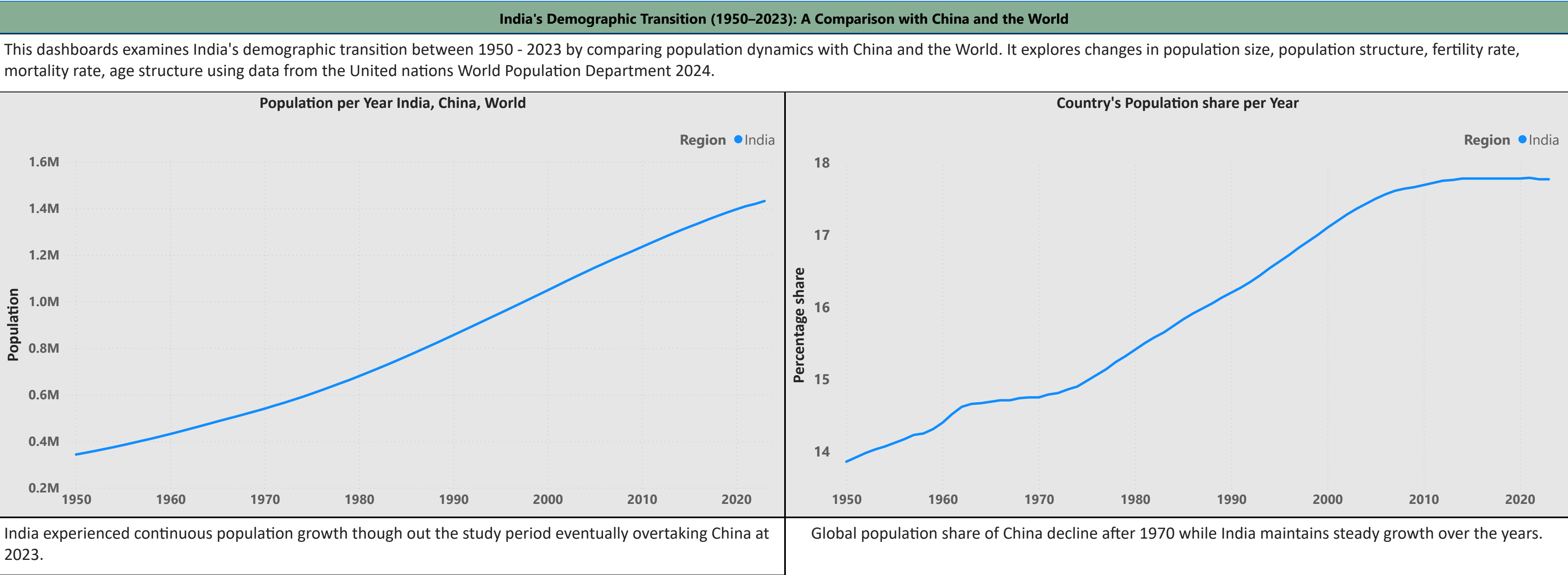
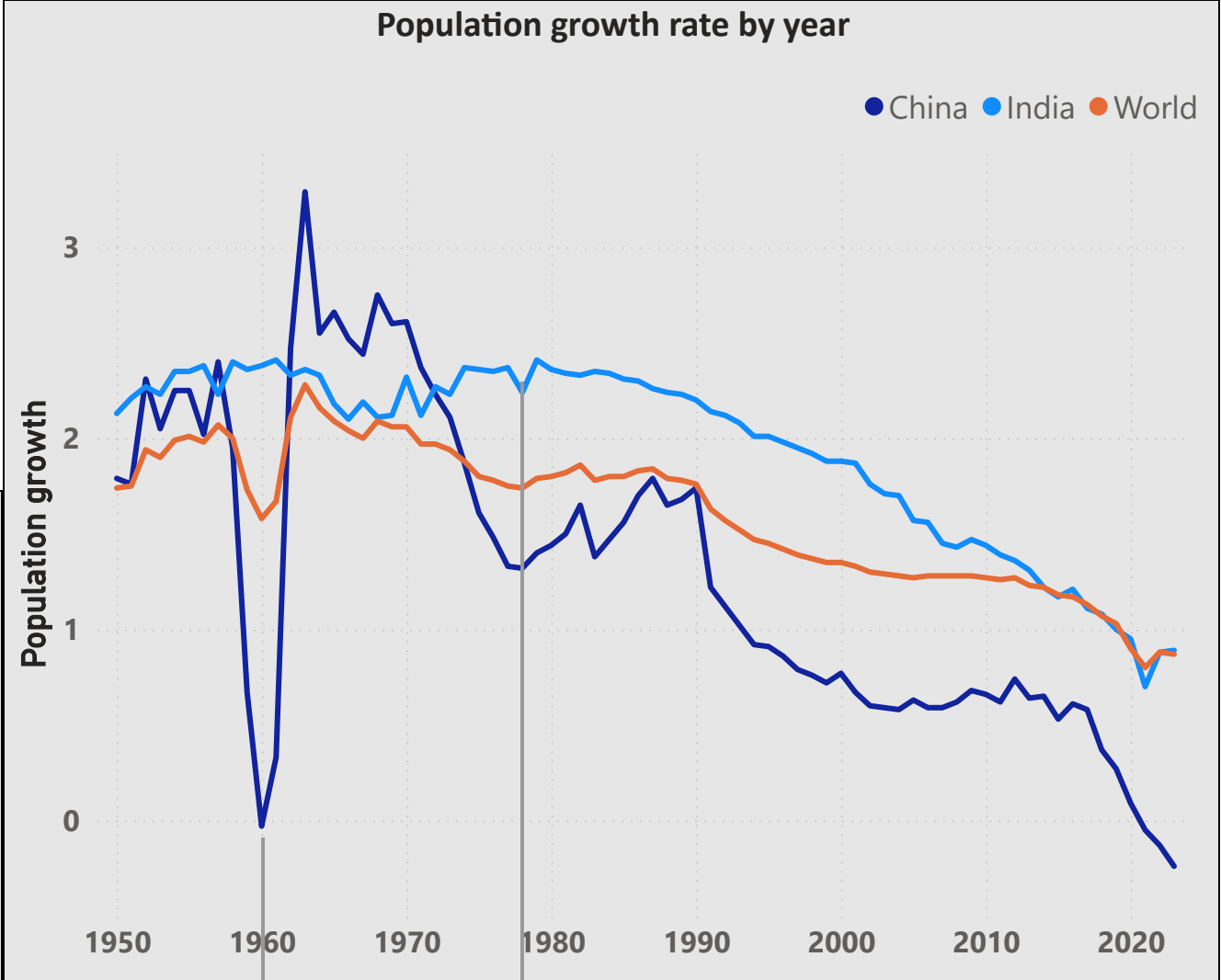
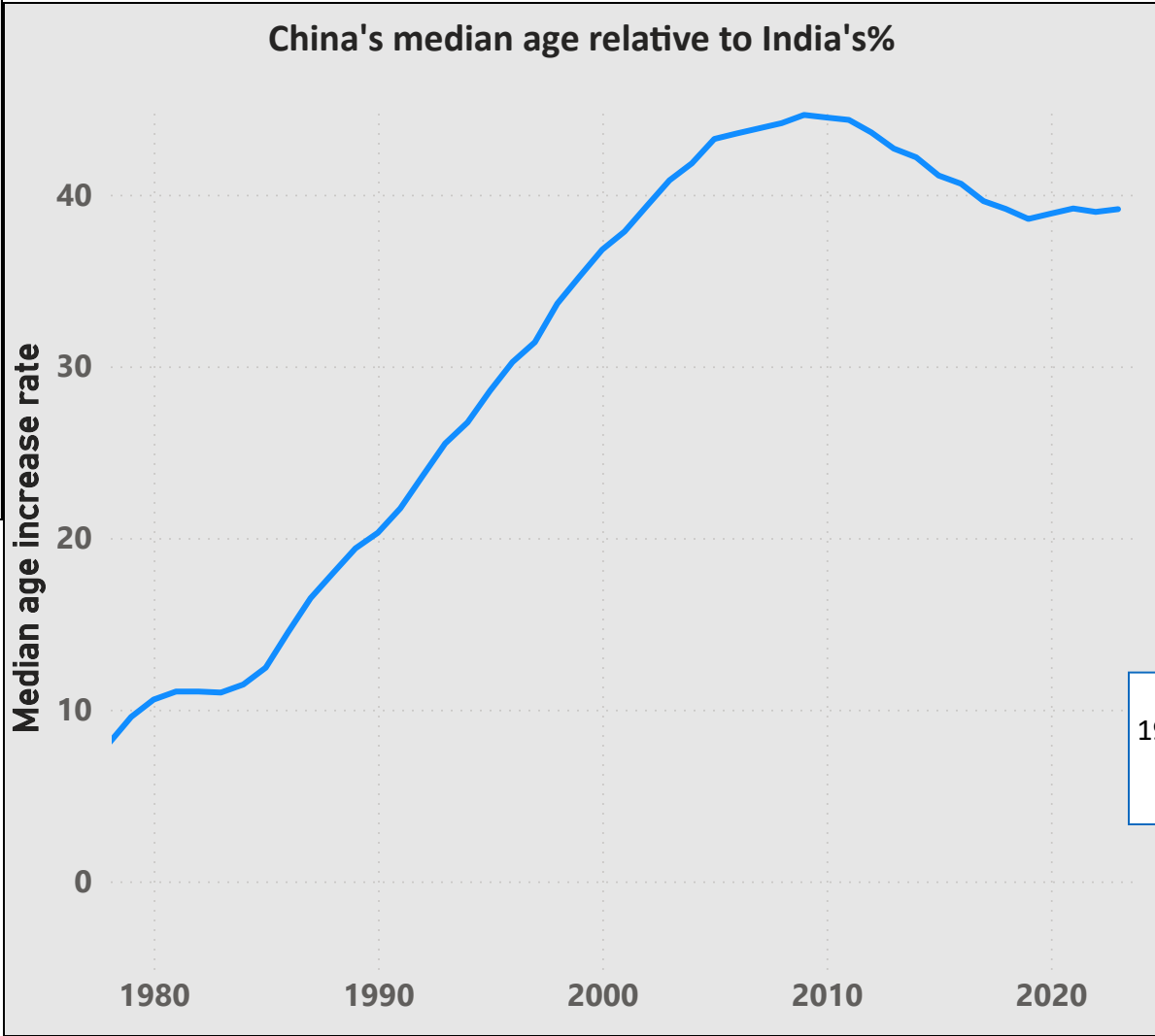
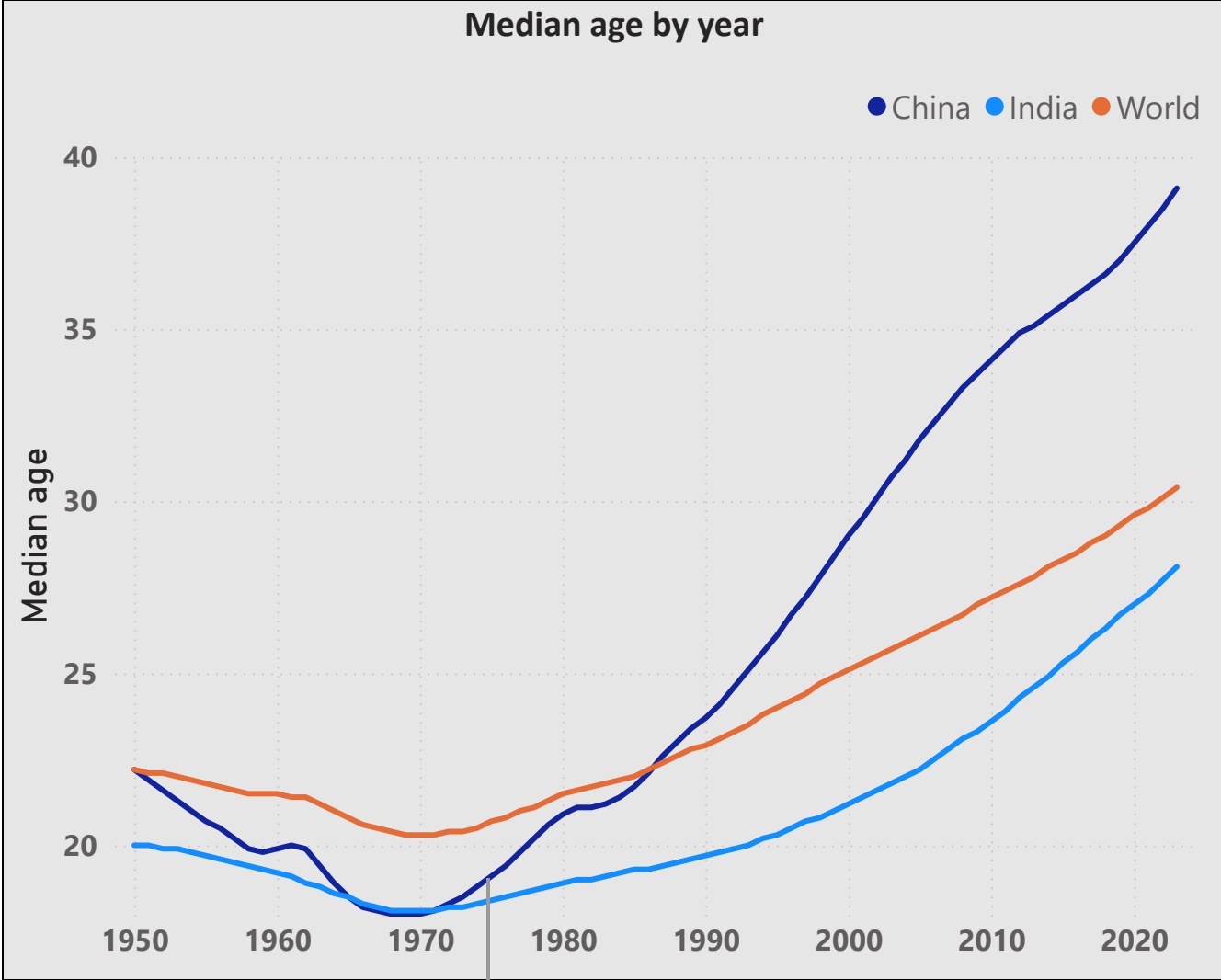




Median age and Population growth rate India with China and the World (1950 - 2023)



Emergency period, India's force sterilization campaign started 1975-1977

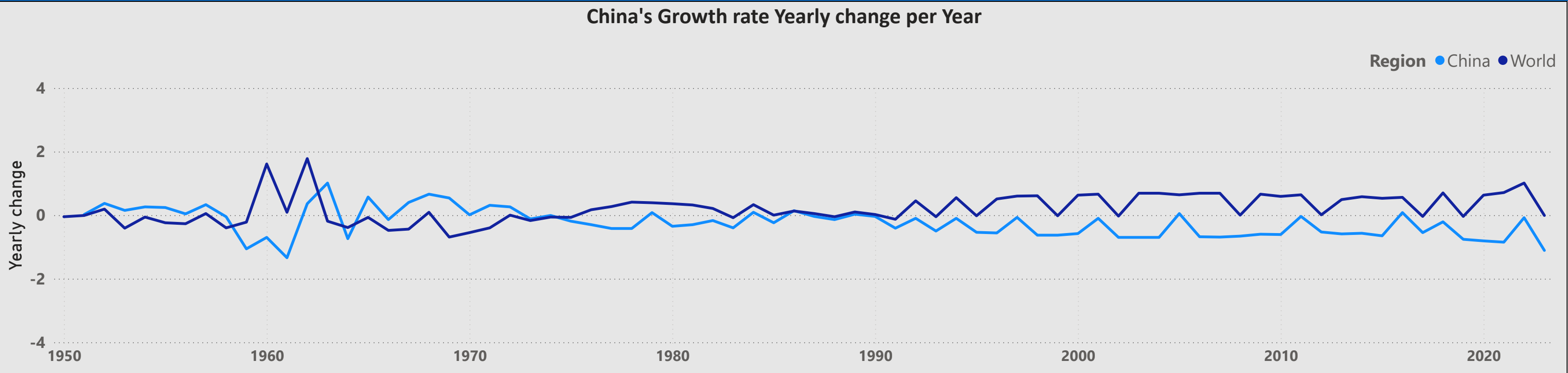
1960, The great Chinese Famine

1977, India's Janata goverment policy legalized minimum marraige age for 18 for females and 21 for males

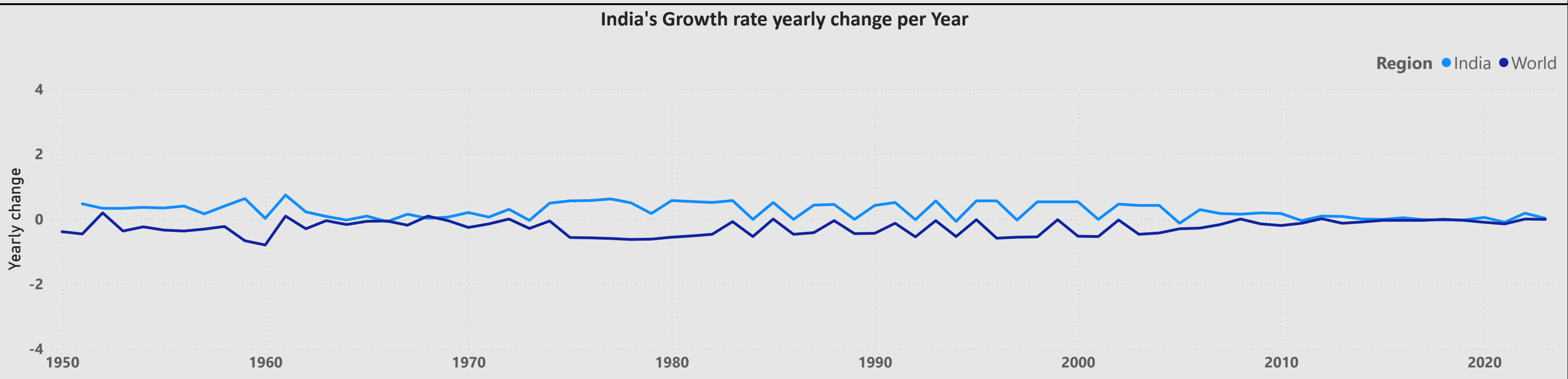
China's median age advantage over India: Lowest median age for India observed at 1970 is 18.00, for China 18.10 and for world 20.30. That's represents a huge number working age population across global alongside with India and China. Chinese median age seems closer to India's till 1979 after that the gap widened substantially coincides China's demographic transition, including implementation of one child's policy by leader Dang Xiaoping at 1979, rapid decline fertility rate and improvement in life expectancy. By around 2010 China's median age was approximately 44.49% higher relative to India's by 1.12% per year. By 2023, China's median age had reached 39.1 years, compared with 28.1 years in India, indicating a substantially older population.

India's steady Population growth rate per year: The high growth rate of population of India about 2.5% on 1960 to till 1994 2.01% has been one of the most persistent development challenges, as the country with merely about 2.4% of the world's lands area, supports disproportionately large share of global population about 15% - 17%, leading immense pressure on resources, infrastructure and public services. While China experienced slowed growth rate over the year which coincides policy implementation of The great leap forward by Mao Zedong leads forced industrialization, poor agricultural cultivations on 1960 keep growth rate below 2.0 per year. Which results high mortality rate and slower population growth.

Yearly change in Growth Rate (1950-2023): India, China vs the World					
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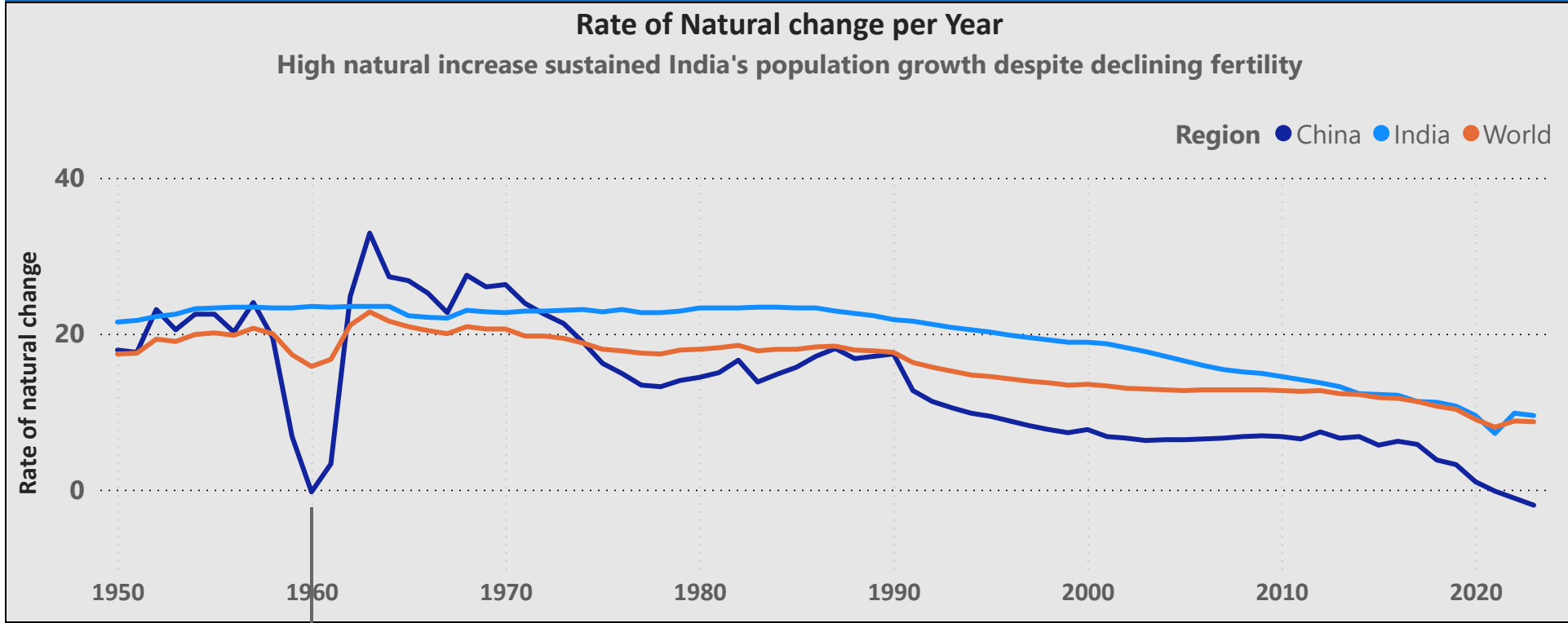


At 1960 China faced abrupt decline on population about -0.70. 1961 it further decrease about -1.34. This period coincides The great leap forward, subsequent famine leads high mortality rate, low birth rate cause low population growth. while global population growth rate comparatively remained stable. This demographic shock contributed to a noticeable change in global trend.

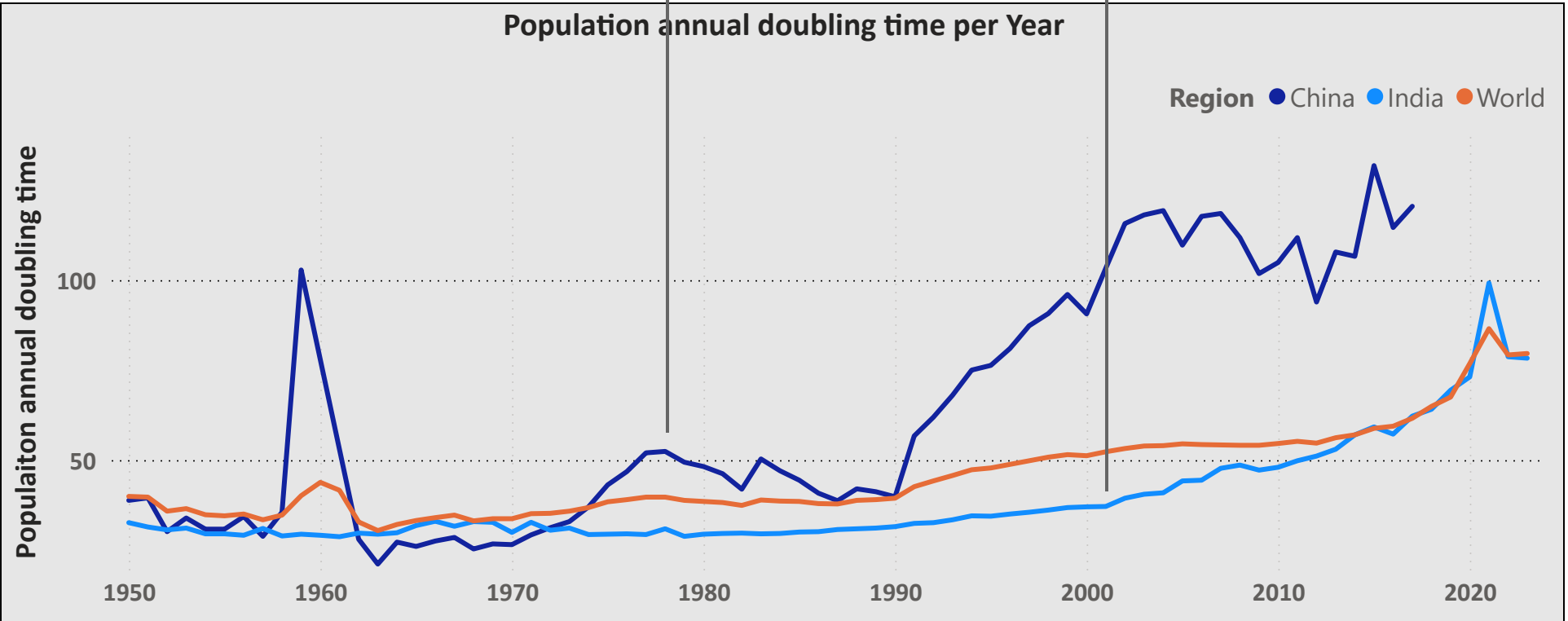


Unlike China India didn't experienced sudden demographic change in population growth during the study. Instead very little demographic changes observe which is relatively small, including gradual demographic transition rather than a sharp demographic shock.

Rate of natural change and annual doubling time per Year (1950 - 2023)



China's The great Leap forward, 1960

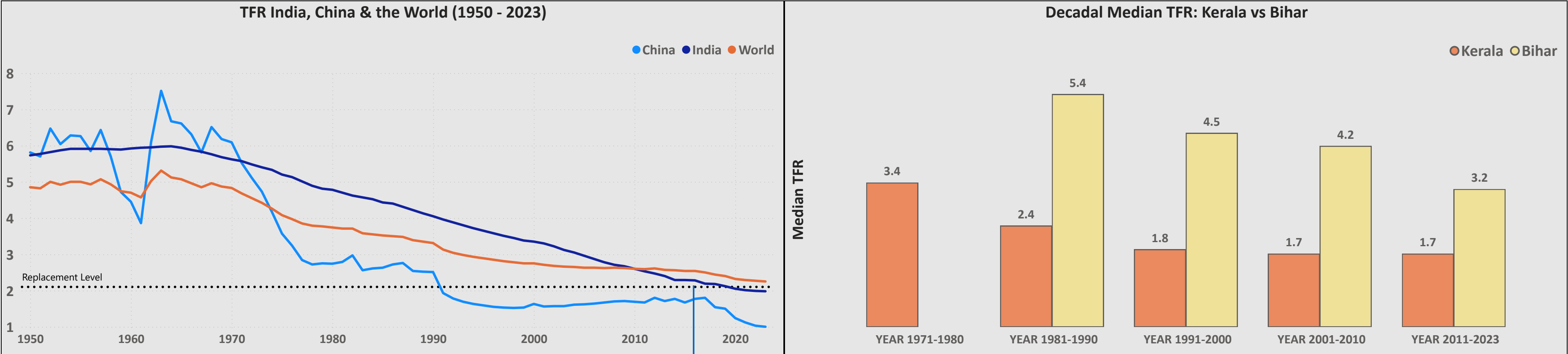


Summary:

Rate of natural change (birth - death): Post independent development in India resulted significant reduce in death rates due to improvement in health infrastructure, but birth rate remain significantly high which kept natural change rate above 20.0% for 45 years after 1950. After natural change rate decline in a steady path with small dip observed on 2020 due world wide covid pandemic reduces natural change rate below 10%. Natural change rate of China lacked India below 22.90% after 1972, the gap substantially wider over the year after 1992 it became 10.50%. Rapid urbanization, health infrastructure, one child policy implementation till 2015 leads lower fertility rate.

Annual Doubling time: India's steady increase in doubling time 1960 from 29.10 years to till 2023 78.30 years. It represents India's population growth slowed year by year dramatically as median age 28 years due to declining fertility and slower growth rate but still with a reasonable advantage compare than China with young working population which can drive India's economic growth. China's population doubling time much higher than India 120.50 years at 2017 which significance lower growth rate and aging population with median age of 36.10 years.

Fertility rate of India in comparision with China and the World (1950 - 2023)



Births in 2023

2.32 Cr

Sex ratio (male/100 female)

107.0

Year

1960

1970

1980

1990

2000

2010

2023

Region

China

India

World

Mission Parivar Vikas MPV, 2016 - access to contraceptives and family planning services at doorstep on high fertility state above 3.0 Bihar, UP, Chhattisgarh, Rajasthan, Jharkhand, Assam, MP

The ministry of health & family welfare leads to Inida's family planning programme. The red triangle represents symbol of family planning, health and contraception services in India.

A family planning stamp 1989

What is the trajectory of Indian population growth?

According to UN reports India surpassed China the most populous country in 2023. Women in reproductive age 15 - 49 have fertility rate 1.9 - A point at which the population can replace itself in one generation. A steady decline on fertility observed at all socio economic groups across India. Kerala is the first state where woman fertility rate decreases below replacement level 2.0 at 1988 when the national fertility was still 4. Bihar maintained highest fertility rate among major states, remaining above 5 births per woman for several decades before declining to 3.2 in 2023. Most of the India's southern and western states, fertility rate reached below replacement level.

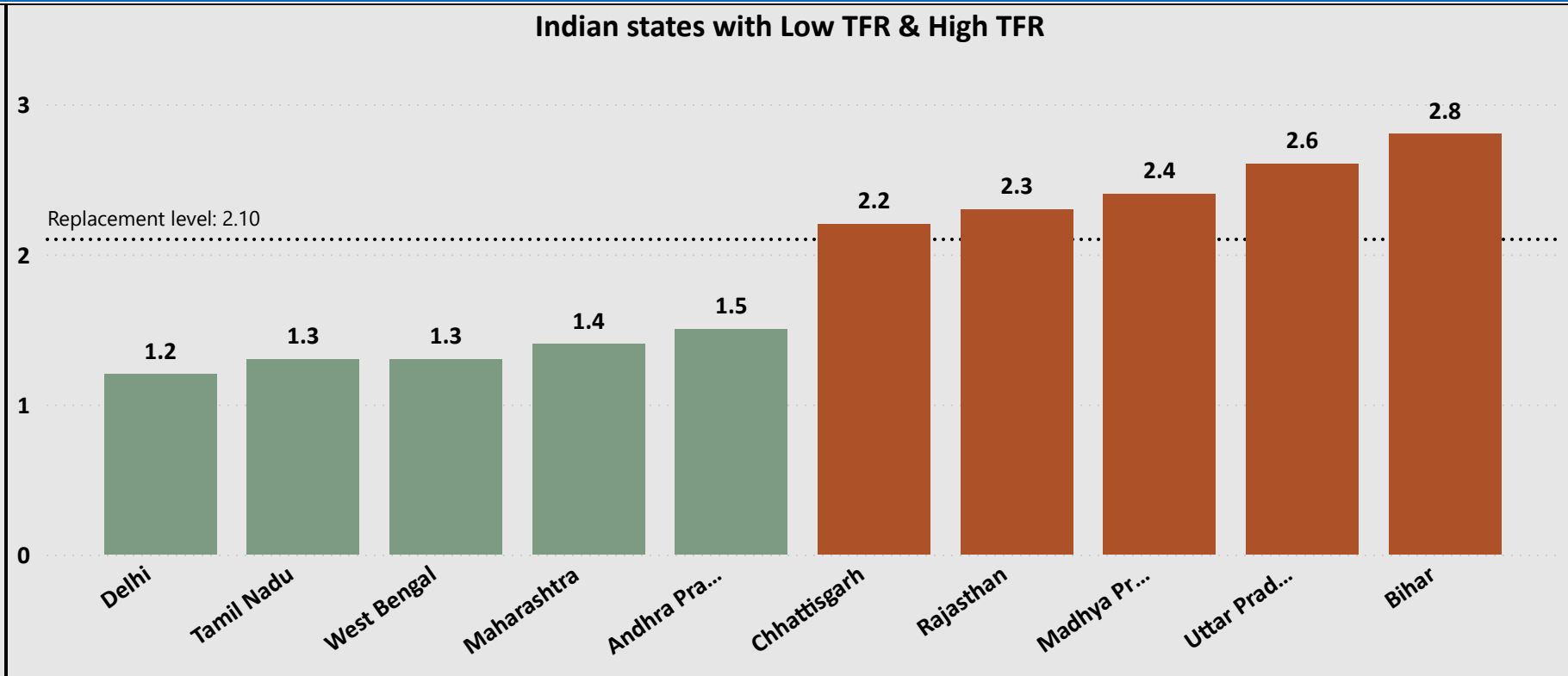
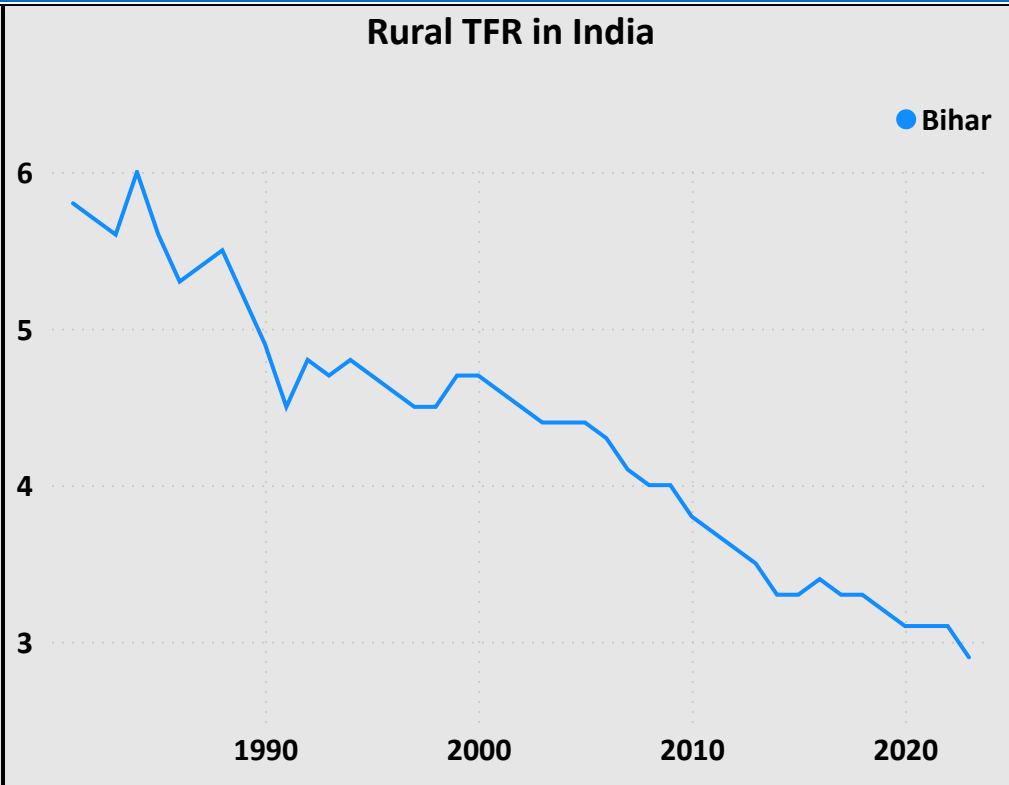
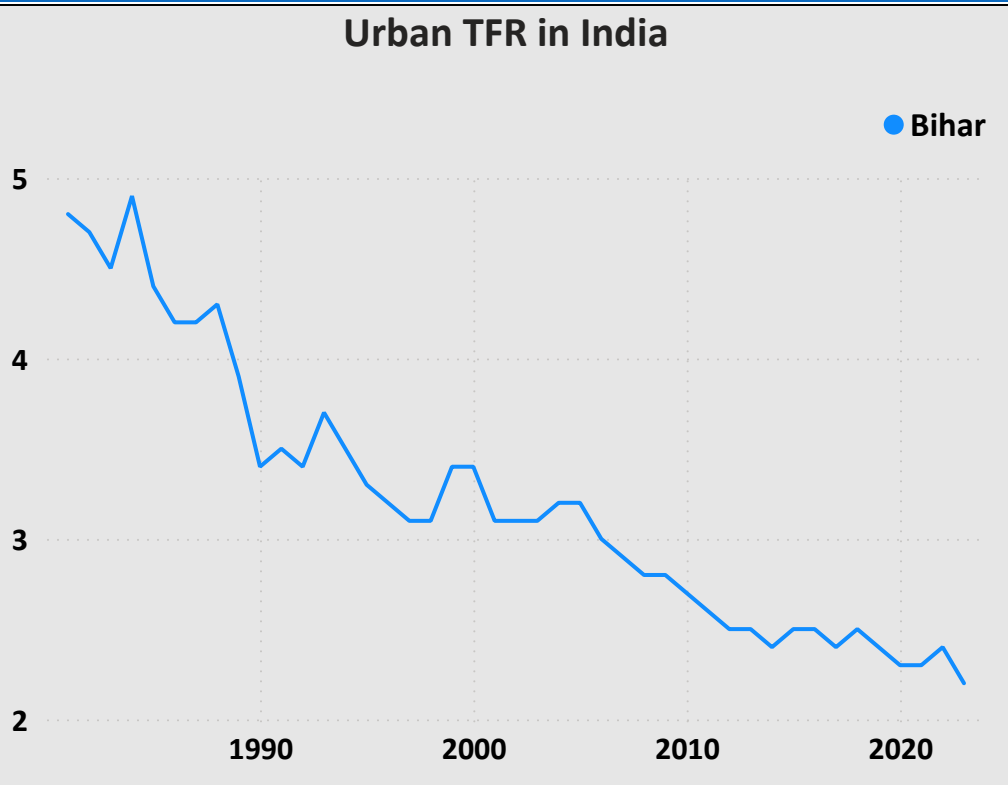
Transition in Fertility rate of INDIA -

After Independence India's fertility rate was around 5.9 during 1950 - 1960, reflects large family size and limited access to healthcare and family planning. By the end of last decade TFR had reduced to around 2.5.

Causation -

- Rising female education, delayed marriages one of the reason of fertility decline in India, educated women tends to prioritize employment.
- Improved child survival reduced precautionary fertility where family earlier had more children due to high child mortality,

Regional trends of fertility rate of India (1950 - 2023)

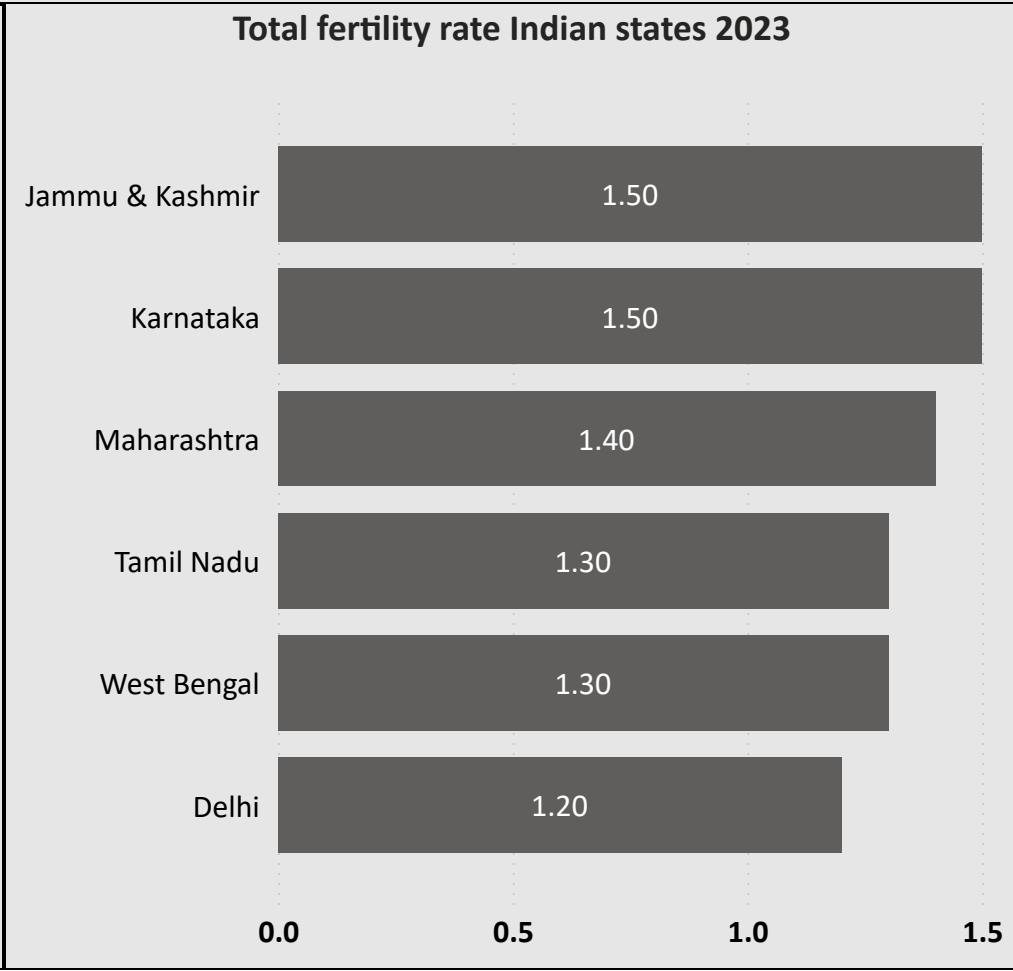
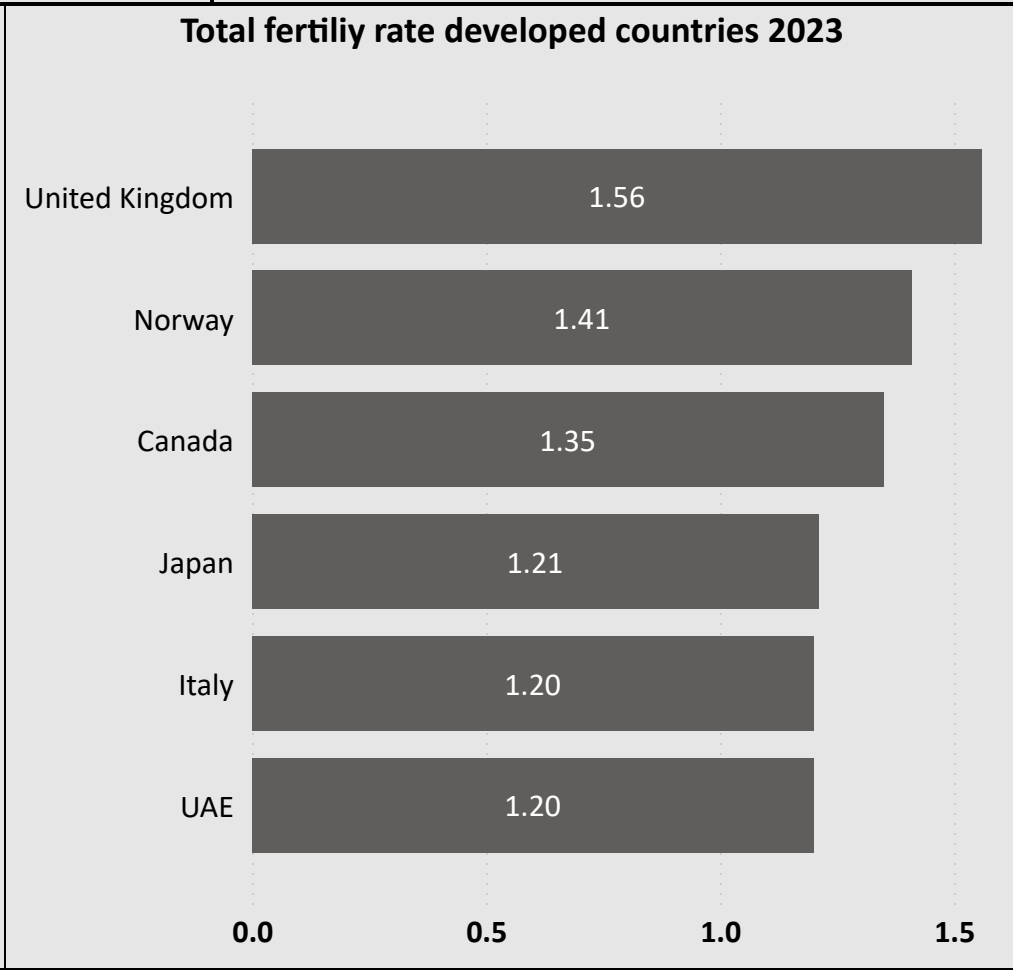


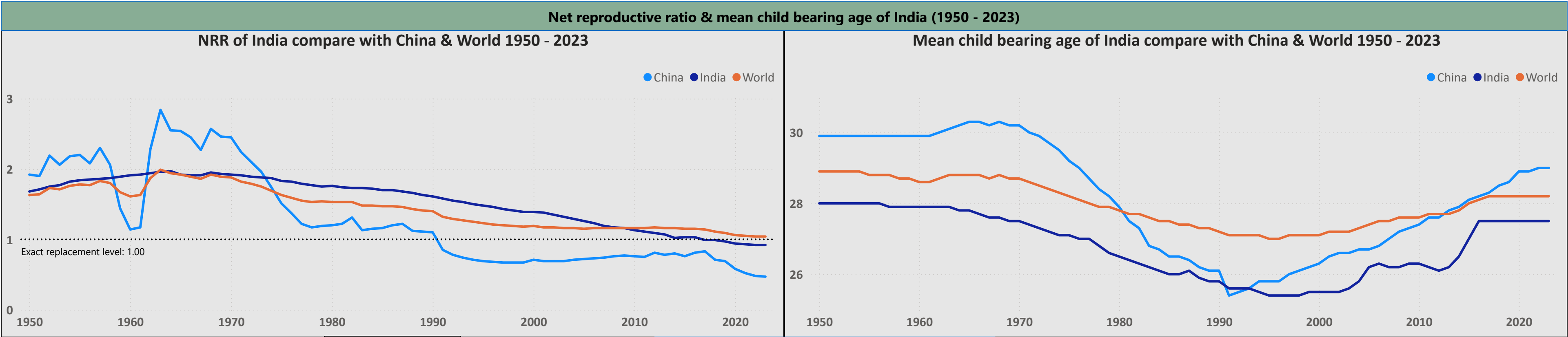
Regional trends : Fertility rate in rural area is remain significantly higher than urban area through out the study period. Urban area reached below replacement level fertility much earlier than rural area, reflecting much faster demographic transition. Urban Madhya Pradesh reached below replacement level fertility (TFR 1.9) in 2020, whereas rural Madhya Pradesh still recorded a TFR in 2.8 in 2020. Tamil Nadu's fertility rate achieved below replacement level at 1993 followed by Andhra Pradesh in 2004, after a decade later. West Bengal represents an interesting exception, despite being government's one of the affluent states, it reached below replacement level at 2005. As per 2023 Bihar, UP, MP, Rajasthan these four Indian states remains above the replacement level among them Bihar recorded highest fertility rate 2.8 births per woman.

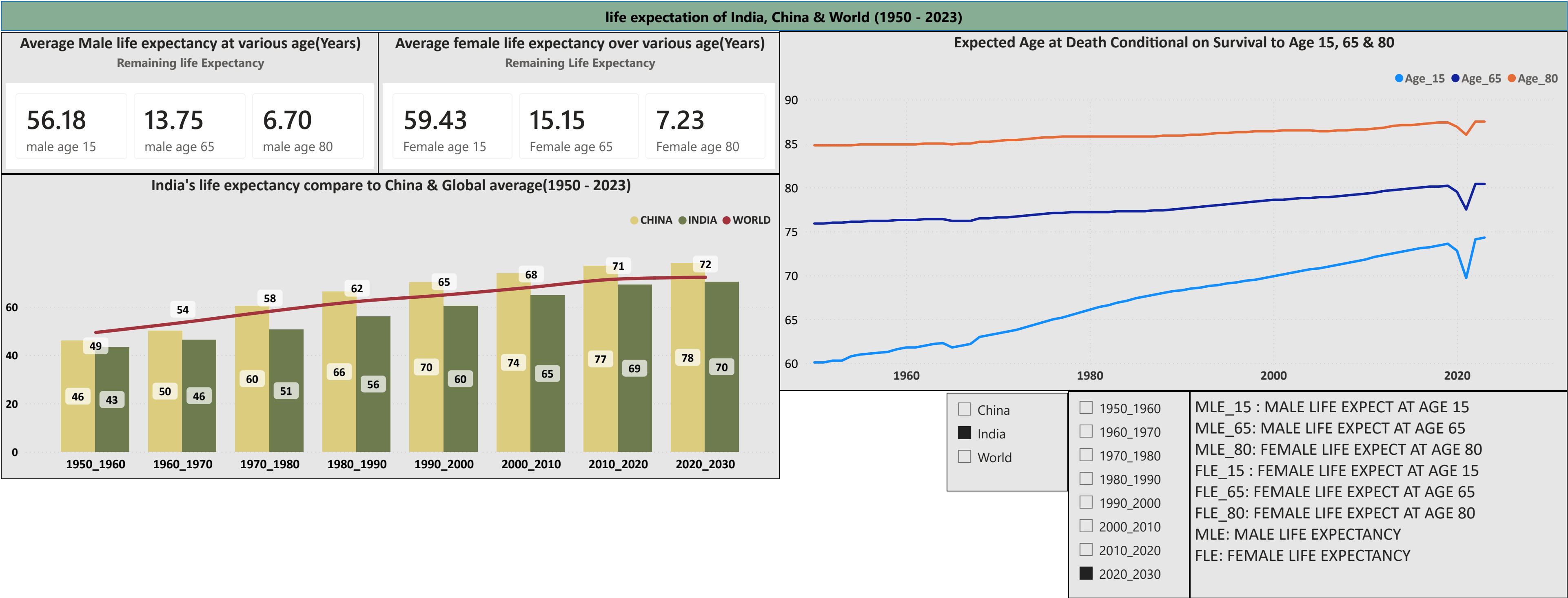
Compare with global trends:
Some of Indian states recorded fertility level as same as developed countries, like Tamil Nadu's fertility rate is similar to Japan and Delhi and West Bengal fertility have levels significantly close to Italy.

Causation - Most of the southern and western region of India recorded replacement level fertility throughout the study period, therefore levels of fertility are generally decline as education, healthcare access, income levels generally improve as in countries in the developed world. In contrast poorer group has been observed higher TFR than educated group, however TFR falls when they get richer and more educated over time. India's rural TFR is still on average 26.93% higher than urban TFR in 2023.

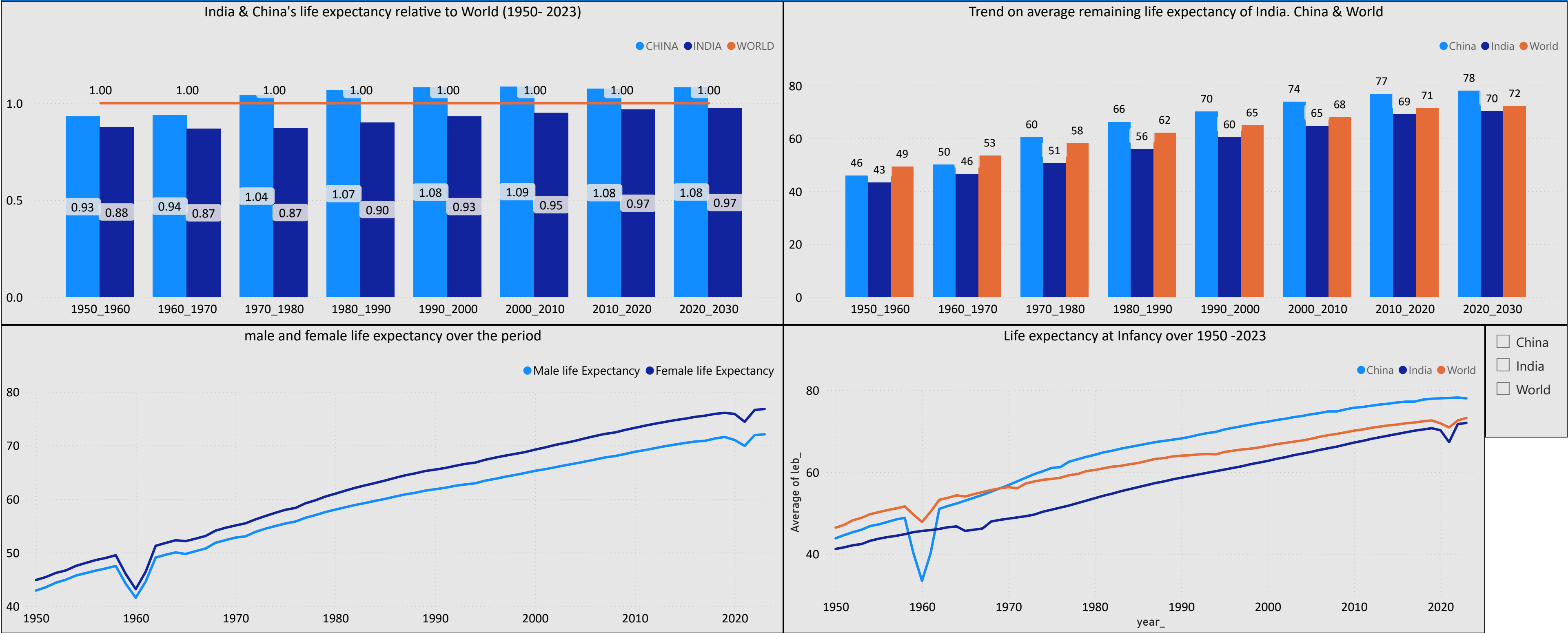
- location
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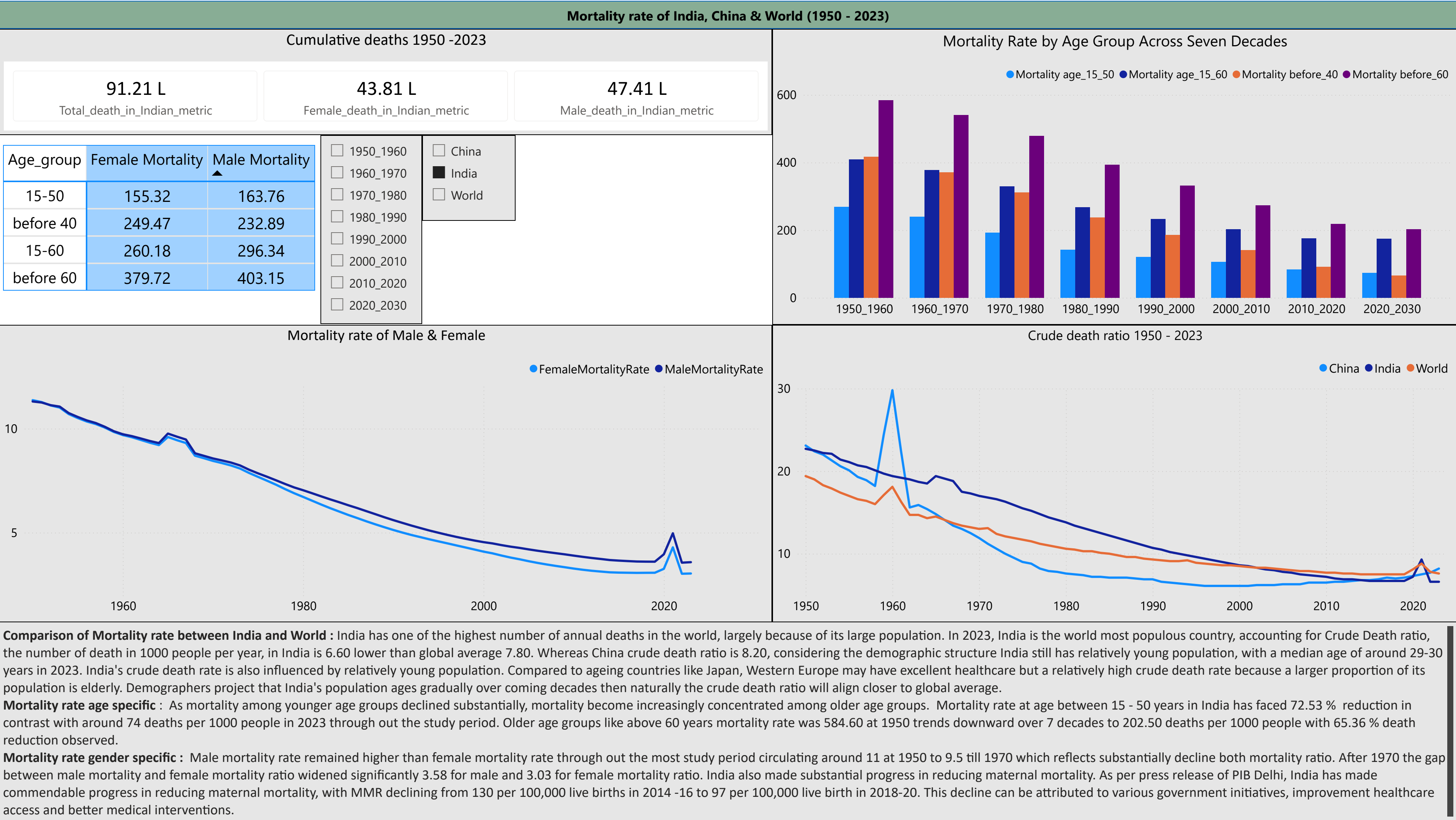
life expectancy at birth of India, China & World (1950 - 2023)



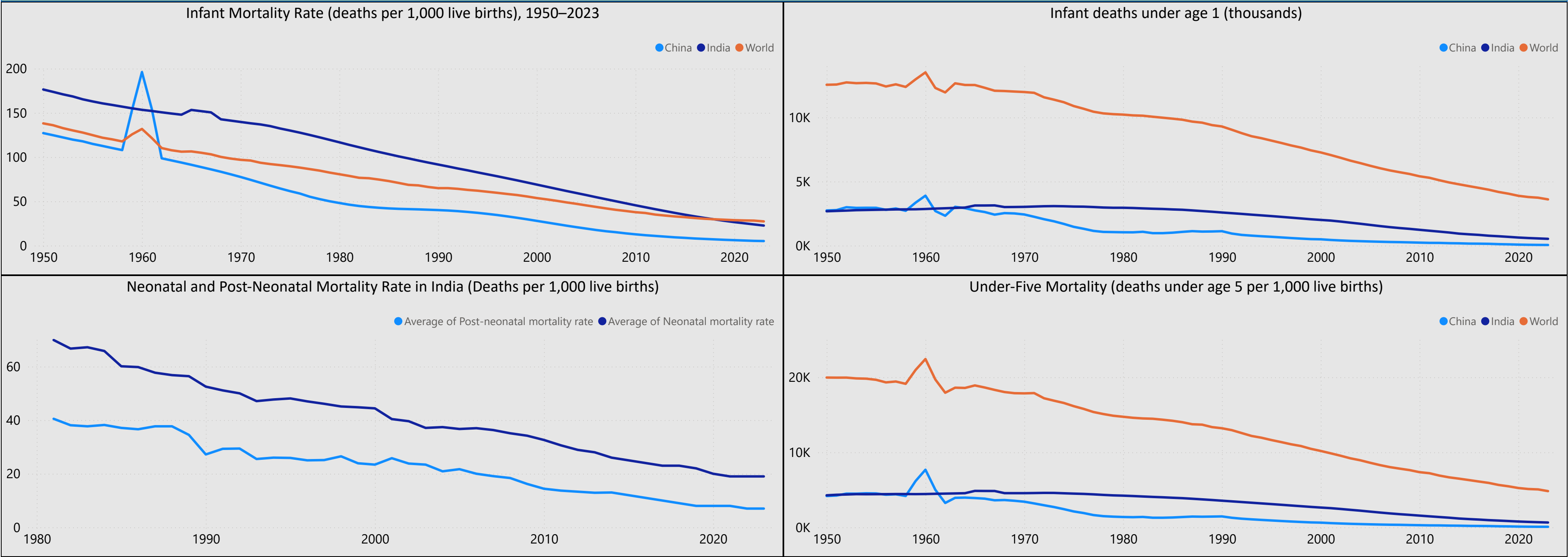
Life expectancy at birth: One of the India's major demographic achievement has been increase in life expectancy over past seven decades. This improvement was supported by declining infant and child mortality, better healthcare, improved nutrition and broader access to education. In 1950 child were born could expect to live only 41 years whereas the global average was 46 years, not only had life expectancy in India gone up to 72 years but also the gap between global and India had also narrowed substantially throughout the study period. Life expectancy has risen steadily in India over time. The only recent time life expectancy of India slid backwards was during the covid19 pandemic at 2020, when life expectancy all other countries fell significantly.

Where does India stand compare to China? China surpassed the World average at 1970, reaching approximately 56 years life expectancy. By 2020, China life expectancy had reached around 78 years, compared with approximately 70 years in India which significance of strong health infrastructure, high education and as a result of rapid urbanization. Throughout the study period India lags slightly differ by on an average 10 years gap from China. But India's direction on life expectancy observed gradually upward over the period.

Female advantages in life expectancy: Across the world it observed that female live longer than man. Over few decades the gap between man and woman life expectancy remain almost identical. whereas for India the scenario was something different before 1980, where female expected to live around 40 years at 1950 which is 2 years less than average male life expectation. But the table has turned in favor of female in 1980 and till present at 2023 female life expectancy increased around 4% than male life expectancy. In present days a newborn girl could expect to live around more than 3 years than a boy born at same year.



Infant Mortality rate of India, China & World (1950 - 2023)



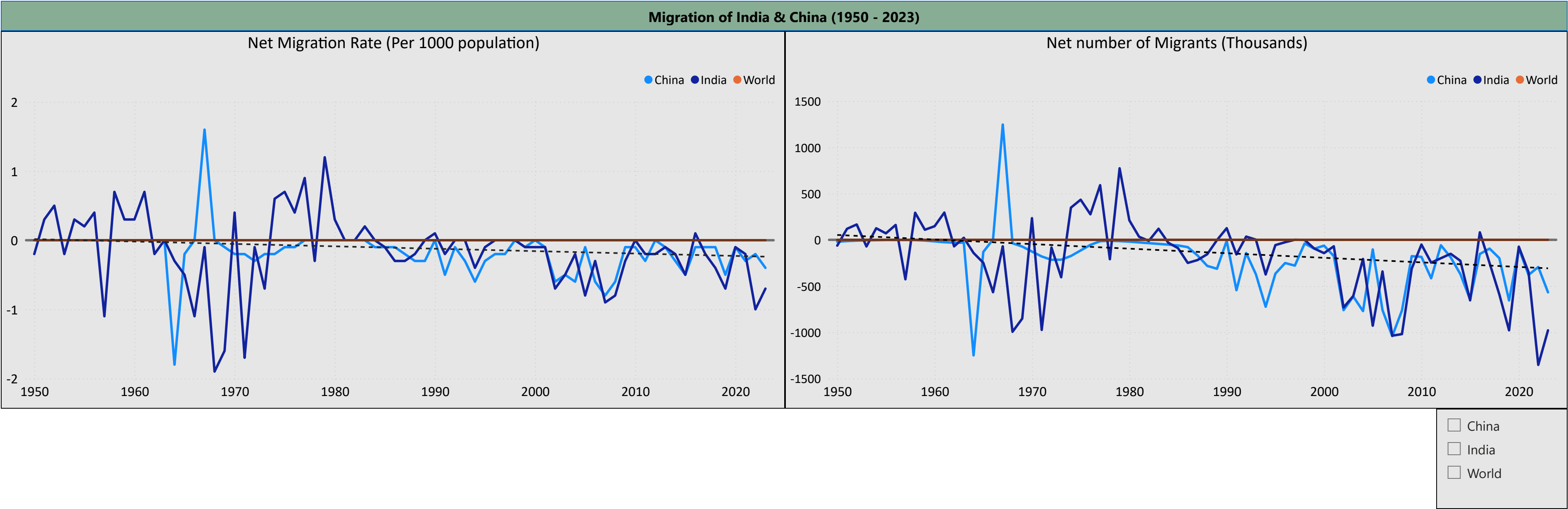
India's Infant Mortality Transition: Over few decades global child survival witnessed a dramatic transformation marked by substantial decline in infant mortality measures by the infant who couldn't survive their first year of life. In 1950 India bore a huge mortality burden approximately 176.40 deaths per 1000 living births, trailing global average by staggering margin of nearly 38 deaths. Over the subsequent decades this gap steadily narrowed until 2017 nearly 22.60 deaths. In this context of demographic transition of mortality rate China maintains its mortality below global average most of the study period and declined to around 5 deaths per 1000 living births by 2023.

When are infants at greatest risk?? As infant mortality decline the contribution of neonatal mortality became increasingly important to relative to post neonatal mortality. As per study Neonatal period, within first month of birth and post neonatal period, from the infants second month to first year are the most vulnerable period. However India has recorded steady decline from approximately 70 neonatal deaths to 19 deaths per 1000 living births over subsequent decades.

Causation of Infant Mortality: As per Cause of Death Statistics 2020-22 released by Office of the Registrar General & Census Commissioner, India; major causes of infant mortality in India are - Prematurity & low birth weight (31.6%), Pneumonia (16.2%), Birth asphyxia & birth trauma (10.2%), Other Non-Communicable Diseases (7.8%), Congenital anomalies (5.3%), Sepsis (4.1%), Fever of unknown origin (3.9%), Diarrheal diseases (3.6%), Injuries (2.8%), Ill-defined or cause unknown (11.5%), and all Other Causes of deaths (2.8%). [Source: National Health Mission]

What are the government initiatives taken for prevent infant deaths?

- National Health Policy (NHP-2017) initiatives.
- Establishment of Sick New Born Care Units at District Hospitals, newborn stabilization Units at Community Health Centers.
- Home Based New Born Care (HBNC) through ASHAs with series of home visits.
- Immunization against seven vaccine preventable diseases.
- Antenatal, Intra-natal and Postnatal care including Iron and Folic Acid supplementation to pregnant & lactating women for prevention and treatment of Anaemia.
- Rashtriva Bal Swasthya Karyakram (RBSK), an introduction of child health screening for 4Ds i.e. defects at birth, deficiencies, diseases, development delays and their management among the children 0-18 years of age.



Migration in India:

From an economic perspective people migrate to other are because of jobs, education, health facilities, business and even marriages. but at its core it is often about labor market and it is driven by two main factors, push and pull factors.

Push factors: lack of jobs, poor living conditions, low wages may one of the reason people leave their hometown or country for finding better opportunity.

Pull factors: better job opportunities, higher wages, hope for better life attracts people to move to a particular place.

India witnessed steady emigration from around 1990 to present years. It is the result of that India's urban areas aren't creating enough jobs. This comes down to Indian are finding much better opportunities abroad. Indians who migrates to another country like USA and UAE - see a massive jump in income. For example low skilled worker can tripled their income in UAE, whereas moving to the US could mean earning five times more.

Causation of Emigration :

The economic advisory Council suggests urban areas no longer seem attractive as they did earlier. Some experts argue India's urbanization is slowing and economic growth is stagnated, making cities less appealing as engines of opportunities.

Government initiatives :

- The Ministry of labor and Employment launched eShram portal. it enables registered workers access to track benefits of multiple schemes directly through eShram.
- Welfare Schemes for the Migrant Laborers, improves the lives of workers by providing access to opportunities for skill development. financial stability and social security.
- Pradhan Mantri Shram Yogi Maan-Dhan Yojana (PMSYM), aims to provide a minimum assured pensions to workers in the unorganized sectors including migrant workers.

Indicator Formula & Methodology

Population growth rate: Population growth rate is the fractional change in the size of population over a specific period.
= (Final population size - Initial population size) * 100 / (initial population * elapsed time period)

Rate of natural change: the rate of natural change is the difference between the number of live births and the number of deaths in a population over giant period.
= Crude Birth rate - Crude Death rate

Population annual doubling time: It is the projected number of years it takes for a given population to double in size, assuming its annual growth rate remains constant.
Rule 70 approximation, Td = 70/r; Td: Doubling time, r = Annual population growth rate expressed as percentage
Exact formula, Td = ln(2)/r; r: Annual growth rate expressed as a decimal

Total fertility rate: TFR is the average number of children a woman have in her lifetime if she experienced the current age specific fertility rates throughout her childbearing years (typically age 15 - 49).
TFR = n * summation of (i = 1, b) ASFR_i
ASFR: Age specific fertility rate for group i = (Birth to woman in age group i) / (Total woman in age group i)
n = width of age group interval, for 5 year age groups, n = 5

Net reproduction rate: NRR is the average number of daughters a newborn girl is expected to have during her lifetime, taking current fertility and female mortality conditions into account.
NRR = summation of (x=15,49) (ASFR_{f,x} * S_x) ; ASFR_{f,x} : age specific birth rate at age x
S_x: probability of newborn female surviving to age x

Crude birth rate: It measures the total number of live births occurring in a given population during a specific year, relative to every 1000 mid year residents.
CBR = (Total Live Births in a Year) * 1000 / (Total Mid year population)

Mean child bearing age: It is the average age of woman at the time they give birth to their children, based on current age specific fertility rates.
MAC = summation of (i = 15-19,45-49) (i, ASFR_i) / summation of (i=15-19,45-49) ASFR_i)

Mortality rate: It is a measure of the frequency of deaths occurring within a defined population over a specified time interval, relative to the population exposed to the risk of deaths.
Mortality rate = (D/P) * 10^k ; D: Total number of deaths recorded in a given timeframe (usually 1 year)
P: Average or mid year population exposed to the risk of dying.
10^k: A standard multiplier (often 1000 for general demographic metrics or 100000 for specific diseases).

Crude death ratio: Total number of deaths occurring in a given population during a specific year, expressed per 1000 mid year residents.
CDR = Total deaths in a year / Total mid year population * 1000

Neonatal & Post neonatal mortality rate: Neonatal and post-neonatal mortality divide the first year of life into two periods with different mortality risks.
Neonatal mortality: deaths during the first 28 days of life.
Post-neonatal mortality: deaths from 28 days to under 1 year.
Infant mortality rate = Neonatal Mortality rate (NMR)+ Post Neonatal Mortality rate (PNMR)

Net migration rate: it measures the difference between the number of people entering a country/region (immigrants) and the number of people leaving (emigrants) over a specific timeframe, expressed per 1000 mid year residents.
Net migration rate = (Immigrants - Emigrants) / (Total Mid year Population) * 1000

Conclusion

- What did I learn about India's demographic transition from 1950–2023?
- After 1950 India's population started to grow 342million to 1.4 billion in 2023. around 40% population was below 15 years old.
 - India's median age was around 18 years in 1970 and increased to around 29 years by 2023, showing a gradual ageing of the population.
 - Indian population growth rate remained high for several decades before gradually declining.
 - India's rate of natural increase declined to below 10 per 1000 population.
 - India's fertility rate began declining after the 1970s and reached replacement level fertility nationally around the early 2020s. Education, urbanization and access to family planning contributed to the transition.
 - Urban area fertility rate reduction is higher than rural area.
 - India's NRR declined over the study period and fell below 1.0, indicating that each generation of woman was no longer replacing itself at the same level.
 - Child bearing age of India is around 28 years in 2023.
 - India's life expectancy increased substantially from around 49 years old to 70 years old.
 - Female life expectancy higher than male life expectancy with woman reaching around 76 years.
 - India's crude death rate reduces to around 6 deaths per 1000 population.
 - Infant mortality rate of India fell below global average at 2018 and reached approximately 22 deaths per 1000 live birth in latest data.
 - India has experienced predominantly negative net migration in recent decades,

Conclusion:

Since independence India shows a significant improvements on demographic transition. improvements in healthcare, sanitation, education and child survival helped reduce mortality and increase life expectancy. Fertility remained high for several decades, which resulted in rapid population growth. Later fertility began to decline and India's population growth rate gradually slowed. The transition has not been uniform across the country. Several southern and western states reached replacement level fertility much earlier, while some northern states continue to have relatively higher fertility. Urban fertility has also declined faster than rural fertility. India is now entering a new demographic phase. The country still a large working age population, but fertility is falling and the population is gradually ageing. This creates an opportunity to benefit from the working age population, while also creating future challenges related to employment, healthcare, social security and ageing. Overall, the data shows that india's demographic transition is the result of changes in fertility, mortality, life expectancy, population structure and migration occurring at different speeds across periods and regions.